DIRAC COHOMOLOGY OF STANDARD MODULES FOR REAL REDUCTIVE GROUPS

Abstract. Let G be a real reductive group. We prove that normalized parabolic induction for G has zero Dirac cohomology whenever the inducing exponent is nonzero. In the special case of standard modules, we showed that it has nonzero Dirac cohomology if and only if it is twisted-elliptic in the sense of Barbasch–Pandžić–Trapa. Both results can be seen as a real-group analogue of the work of Chan in the setup of graded affine Hecke algebras. The same technique can be applied to get a simplified proof of the generalized vanishing theorem for induced graded affine Hecke algebra modules.

1. Introduction

Let $G = \mathbf{G}(\mathbb{R})$, where $\mathbf{G}$ is a connected reductive linear algebraic group over $\mathbb{R}$. Fix a Cartan involution θ, put $K = G^\theta$, and write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ for the complexified Cartan decomposition. If X is a Harish-Chandra module and S is a spin module for the Clifford algebra of $\mathfrak{p}$, its *Dirac cohomology* is

$$H_D(X) = \ker D/(\ker D \cap \operatorname{im} D),$$

where D is the Dirac operator on $X \otimes S$ defined in Section 3. The fundamental theorem of Huang–Pandžić [8, Theorem 2.3] proves that, for an irreducible module with nonzero Dirac cohomology, the highest weight of any compact spin-cover constituent in $H_D(X)$ determines the infinitesimal character of X.

For graded affine Hecke algebras with geometric parameters, Chan [4, Theorem 6.4] proves that every non-tempered geometric standard module has zero Dirac cohomology. The main purpose of this paper is to establish the analogous vanishing theorem for standard modules of real reductive groups. Indeed, we prove a more general assertion for normalized parabolic induction, without restrictions on the inducing representation beyond admissibility.

Let $P = MAN$ be a real parabolic subgroup with Langlands decomposition relative to θ, and put $\mathfrak{a} = \operatorname{Lie}(A) \otimes_{\mathbb{R}} \mathbb{C}$. For a smooth admissible representation σ of M and $\nu \in \mathfrak{a}^*$, write

$$I_P(\sigma, \nu) = \operatorname{Ind}_P^G(\sigma \otimes e^\nu)$$

for normalized parabolic induction, taking the K-finite vectors when considering its Dirac cohomology.

Theorem 1.1. *Let $P = MAN$ be a parabolic subgroup of G as above. For every smooth admissible representation σ of M and every $\nu \in \mathfrak{a}^* \setminus \{0\}$,*

$$H_D(I_P(\sigma, \nu)) = 0.$$

Now we apply Theorem 1.1 to *standard modules*, which has the form form

$$Y = I_P(\sigma, \nu) = \operatorname{Ind}_P^G(\sigma \otimes e^\nu), \qquad P = MAN,$$

where M admits discrete series, σ is a discrete series or limit of discrete series representation of M, and $\nu \in \mathfrak{a}^*$ has real part in the closed positive chamber determined by P; see Definition 3.3. By Theorem 1.1, $H_D(Y) \neq 0$ implies that $\nu = 0$ and Y is irreducible and tempered with real infinitesimal character.

A $(\mathfrak{g}, K)$-module is called *twisted-elliptic* if it admits an action of the Cartan involution (i.e. a $(\mathfrak{g}, K^+)$-module given in Subsection 3.2 below) with nonzero twisted Dirac index ([3, §1.4]). The second result of this paper is the real-group analogue of Chan's criterion [4, Theorem 6.4 and Corollary 6.6] given as follows:

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Theorem 1.2. *Let $G = \mathbf{G}(\mathbb{R})$, where $\mathbf{G}$ is a connected reductive linear algebraic group over $\mathbb{R}$, and let Y be a standard $(\mathfrak{g}, K)$-module. Then*

(1.1) $H_D(Y) \neq 0 \iff$ *Y is twisted-elliptic tempered (in the sense of Theorem 3.4 below).*

Indeed, the techniques of proving Theorem 1.1 is also applicable to graded affine Hecke algebra setting. This gives an alternative proof of the non-tempered part of [4, Theorem 6.4]. We will present its proof in Section 4 below.

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2. An algebraic vanishing criterion

Let $\mathcal{V}$ be a complex vector space and let $D \in \mathrm{End}_{\mathbb{C}}(\mathcal{V})$. Define

$$\mathcal{H}_D(\mathcal{V}) = \ker D/(\ker D \cap \operatorname{im} D).$$

Lemma 2.1. *Suppose that $C, T \in \mathrm{End}_{\mathbb{C}}(\mathcal{V})$ satisfy*

$$DC + CD = -2T, \qquad DT = TD,$$

and that T is invertible. Then $\mathcal{H}_D(\mathcal{V}) = 0$. In particular, this holds if $DC + CD = -2a\,\mathrm{id}_{\mathcal{V}}$ for some $a \in \mathbb{C} \setminus \{0\}$.

Proof. Since T^{-1} commutes with D, the operator $Q = -\frac{1}{2}CT^{-1}$ satisfies

$$DQ + QD = -\tfrac{1}{2}(DC + CD)T^{-1} = \mathrm{id}_{\mathcal{V}}\,.$$

For every $x \in \ker D$ we therefore have $x = DQx$. Thus $\ker D \subseteq \operatorname{im} D$. The scalar assertion follows by taking $T = a\,\mathrm{id}_{\mathcal{V}}$. □

3. Real reductive groups

3.1. **Dirac cohomology and the infinitesimal character.** Let $\mathbf{G}$ be a connected reductive linear algebraic group over $\mathbb{R}$, and put $G = \mathbf{G}(\mathbb{R})$. Fix a Cartan involution θ and write

$$K = G^{\theta}, \qquad \mathfrak{g}_0 = \mathfrak{k}_0 \oplus \mathfrak{p}_0.$$

The subscript 0 denotes a real Lie algebra, and its omission denotes complexification. The group G need not be connected. Choose an $\mathrm{Ad}(G)$-invariant and θ-invariant symmetric bilinear form B such that

$$B_\theta(U, V) = -B(U, \theta V)$$

is positive definite on $\mathfrak{g}_0$. Extend B complex linearly.

Let $C(\mathfrak{p})$ be the complex Clifford algebra with defining relation

$$c(U)c(V) + c(V)c(U) = -2B(U, V)\,\mathrm{id}\,.$$

Fix an irreducible $C(\mathfrak{p})$-module S. Let $\widetilde{K}$ be the pullback of $\mathrm{Pin}(\mathfrak{p}_0) \to O(\mathfrak{p}_0)$ under $\mathrm{Ad}\,|_{\mathfrak{p}_0} : K \to O(\mathfrak{p}_0)$. For a Harish-Chandra $(\mathfrak{g}, K)$-module X, define the *Dirac operator* $D_X : X \otimes S \to X \otimes S$ by:

$$D_X = \sum_{j=1}^{\dim \mathfrak{p}_0} \pi_X(Z_j) \otimes c(Z_j)$$

where (Z_j) is any B-orthonormal basis of $\mathfrak{p}_0$. The *Dirac cohomology* of X is

$$H_D(X) = \ker D_X/(\ker D_X \cap \operatorname{im} D_X),$$

Fix a maximal torus T_0 of the identity component K° of $K.$, with Lie algebra $\mathfrak{t}_0$. Its centralizer $\mathfrak{h}_0 = \mathfrak{t}_0 \oplus \mathfrak{a}_0$ in $\mathfrak{g}_0$ is a fundamental θ-stable Cartan subalgebra; here $\mathfrak{a}_0 = \mathfrak{h}_0 \cap \mathfrak{p}_0$. Fix positive roots for $(\mathfrak{k}, \mathfrak{t})$ and let $\rho_c \in \mathfrak{t}^*$ be their half-sum. Regard $\mathfrak{t}^*$ as a subspace of $\mathfrak{h}^*$ by extension by zero on $\mathfrak{a}$. An infinitesimal character is called *real* if its representative in $\mathfrak{h}^*$ lies in $i\mathfrak{t}_0^* \oplus \mathfrak{a}_0^*$. This condition is Weyl invariant and independent of the chosen θ-stable Cartan; see [6, pp. 43–44].

Theorem 3.1 (Huang–Pandžić)**.** *Let X be an irreducible Harish-Chandra module with infinitesimal character represented by $\Lambda \in \mathfrak{h}^*$. If an irreducible $\widetilde{K}^\circ$-constituent of $H_D(X)$ has highest weight γ, then*

$$\Lambda \in W(\mathfrak{g}, \mathfrak{h})(\gamma + \rho_c),$$

where $W(\mathfrak{g}, \mathfrak{h})$ is the Weyl group of $(\mathfrak{g}, \mathfrak{h})$. In particular, X has real infinitesimal character.

The algebraic identity underlying this theorem is [8, Corollary 3.5], with the disconnected formulation in [6, Theorem 2.1]. Let $\Delta : U(\mathfrak{k}) \to U(\mathfrak{g}) \otimes C(\mathfrak{p})$ be the diagonal embedding defined by the inclusion $\mathfrak{k} \subseteq \mathfrak{g}$ and the infinitesimal spin action on $C(\mathfrak{p})$. For every $z \in Z(U(\mathfrak{g}))$, there is an element a_z of $U(\mathfrak{g}) \otimes C(\mathfrak{p})$ such that

$$z \otimes 1 - \Delta(\zeta(z)) = Da_z + a_z D,$$

where $\zeta : Z(U(\mathfrak{g})) \to Z(U(\mathfrak{k}))^K$ is the homomorphism which, under the Harish-Chandra isomorphisms, is restriction from $\mathfrak{h}^*$ to $\mathfrak{t}^*$ [8, Theorem 5.5].

3.2. **Twisted Dirac index.** We now introduce the notion of twisted Dirac index as given in [3]. Write

$$G^+ = G \rtimes \langle \vartheta \rangle, \qquad \vartheta^2 = 1, \qquad \vartheta g \vartheta^{-1} = \theta(g), \qquad K^+ = K \times \langle \vartheta \rangle.$$

Following [3, §1.4], a $(\mathfrak{g}, K^+)$-module (π, X) is a $(\mathfrak{g}, K)$-module with an additional action $\pi(\vartheta)$ satisfying $\pi(\vartheta)^2 = 1$ and

$$\pi(\vartheta)\pi(k)\pi(\vartheta) = \pi(k) \quad (k \in K),$$
$$\pi(\vartheta)\pi(Z)\pi(\vartheta) = \pi(\theta Z) \quad (Z \in \mathfrak{g}).$$

This definition applies in both the equal-rank and unequal-rank cases. Write $X^\pm$ for the ± 1-eigenspaces of $\pi(\vartheta)$. Since $\theta|_{\mathfrak{p}} = -1$, the operator $\pi(\vartheta) \otimes 1_S$ anticommutes with D_X and therefore acts on $H_D(X)$. Denote its eigenspaces by $H_D(X)^\pm$.

Definition 3.2. For a Harish-Chandra $(\mathfrak{g}, K^+)$-module X with infinitesimal character, the *twisted Dirac index* is

$$I_\theta(X) = [H_D(X)^+] - [H_D(X)^-] \in R(\widetilde{K}),$$

where $R(\widetilde{K})$ is the Grothendieck group of finite-dimensional representations of $\widetilde{K}$. The module X is *twisted-elliptic* if $I_\theta(X) \neq 0$.

By [3, Proposition 1.8 and equation (1.13)],

$$I_\theta(X) = [X^+ \otimes S] - [X^- \otimes S],$$

where the right-hand side is interpreted in the Grothendieck group of admissible $\widetilde{K}$-modules.

In the case when $\operatorname{rank} \mathfrak{g} = \operatorname{rank} \mathfrak{k}$, there is a more notion of the *ordinary Dirac index*, which we now recall its definition and its relation with the twisted Dirac index. In such a case, $\dim \mathfrak{p}$ is even and $S = S^+ \oplus S^-$ has its half-spin grading. Let Γ act by ± 1 on $S^\pm$, and put

$$\widetilde{K}_{\mathrm{ev}} = \{(k, u) \in \widetilde{K} : u \in \mathrm{Spin}(\mathfrak{p}_0)\}.$$

For a Harish-Chandra module Y with infinitesimal character, write $H_D(Y)^\pm_{\mathrm{spin}}$ for the eigenspaces of $1 \otimes \Gamma$ on $H_D(Y)$. The *ordinary Dirac index* is

$$I_D(Y) = [H_D(Y)^+_{\mathrm{spin}}] - [H_D(Y)^-_{\mathrm{spin}}] = [Y \otimes S^+] - [Y \otimes S^-] \in R(\widetilde{K}_{\mathrm{ev}}),$$

where the second equality is interpreted initially in the Grothendieck group of admissible representations. If K is connected, then $\widetilde{K}_{\mathrm{ev}} = \widetilde{K}$.

In the equal-rank case, $\theta = \mathrm{Ad}(k_0)$ is inner. For the extension X of Y and the lift $\widetilde{k}_0 \in \widetilde{K}_{\mathrm{ev}}$ with the normalization used in [3, §1.4], the index characters satisfy

$$\chi_{I_\theta(X)}(\widetilde{k}) = \chi_{I_D(Y)}(\widetilde{k}_0 \widetilde{k}) \qquad (\widetilde{k} \in \widetilde{K}_{\mathrm{ev}}). \tag{3.1}$$

Here χ denotes the character of a virtual representation. Thus, for this extension, the twisted index recovers the ordinary index.

One of the main goals of this paper is to study the Dirac cohomology of standard $(\mathfrak{g}, K)$-modules, whose definition is given by:

Definition 3.3. A *standard* $(\mathfrak{g}, K)$*-module* is of the normalized induced representation

$$Y = I_P(\sigma, \nu) = \mathrm{Ind}_P^G(\sigma \otimes e^\nu), \qquad P = MAN,$$

where σ is a limit of discrete series of M, $\langle \mathrm{Re}\,\nu, \alpha^\vee \rangle > 0$ for every positive restricted root α occurring in $\mathfrak{n}$, and satisfies a certain "parity condition" given in [1].

We identify $Y = I_P(\sigma, \nu)$ with the notation of [3, Definition 5.5]. Choose a compact Cartan subalgebra $\mathfrak{t}$ of $\mathfrak{m}$ and a θ-stable Borel subalgebra $\mathfrak{b}_M \subseteq \mathfrak{m}$ containing $\mathfrak{t}$ which gives a cohomological-induction realization of σ. Put

$$\mathfrak{h} = \mathfrak{t} \oplus \mathfrak{a}, \qquad \mathfrak{b} = (\mathfrak{b}_M \oplus \mathfrak{a}) \oplus \mathfrak{n} = \mathfrak{h} \oplus \mathfrak{u},$$

and let $H = TA$ be the corresponding Cartan subgroup, with $T = H \cap K$. The inducing data determine a one-dimensional $(\mathfrak{h}, T)$-module $\mathbb{C}_\lambda$. With the normalization of [3, Definition 5.5], induction in stages gives

$$Y \simeq X_{\mathfrak{b}}(\lambda) = \mathcal{L}_s(\mathbb{C}_\lambda), \qquad s = \dim(\mathfrak{u} \cap \mathfrak{k}).$$

Note that the Borel subalgebra $\mathfrak{b}$ need not be θ-stable. The θ-twist Y^θ has the same K-action and $\pi_{Y^\theta}(Z) = \pi_Y(\theta Z)$, so $Y^\theta \simeq X_{\theta\mathfrak{b}}(\theta\lambda)$. Following [3, §5.2], exactly one of the following two cases applies to Y.

(i) $Y \cong Y^\theta$. Then Y has two ways of extending into a $(\mathfrak{g}, K^+)$-modules, labeled as Y^ε.

(ii) $Y^\theta \ncong Y$. Then $\widetilde{Y} := Y \oplus Y^\theta$ is a $(\mathfrak{g}, K^+)$-module with $\pi(\vartheta)(x, y) = (y, x)$.

Theorem 3.4 ([3, Theorem 5.7])**.** *Let Y be a standard $(\mathfrak{g}, K)$-module.*

(i) *Suppose that $Y^\theta \simeq Y$, and let Y^ε, $\varepsilon = \pm 1$, be its two standard extensions. Then $I_\theta(Y^\varepsilon) \neq 0$ if and only if Y admits a realization*

$$Y \simeq X_{\mathfrak{b}}(\lambda), \qquad \theta\mathfrak{b} = \mathfrak{b}, \qquad \theta\lambda = \lambda.$$

(ii) *Suppose that $Y^\theta \not\simeq Y$. For $\widetilde{Y} = Y \oplus Y^\theta$ with ϑ interchanging the summands, one has*

$$I_\theta(\widetilde{Y}) = 0.$$

We call a standard module Y twisted elliptic *if it satisfies the hypotheses of (i) above.*

We remark that in the equal-rank case, the above result can also be obtained using elliptic-pairing of $(\mathfrak{g}, K)$-modules studied in [7].

3.3. **The Dirac family.** Let $P = MAN$ be a real parabolic subgroup, with Langlands decomposition relative to θ. Throughout this subsection, write

$$\mathfrak{m}_0 = \mathrm{Lie}(M), \qquad \mathfrak{a}_0 = \mathrm{Lie}(A), \qquad \mathfrak{n}_0 = \mathrm{Lie}(N).$$

Thus $\mathfrak{a}_0$ now denotes the split component of the inducing parabolic; it need not be the split part of the fundamental Cartan used in Theorem 3.1. Put

$$\rho_P(H) = \frac{1}{2}\,\mathrm{tr}(\mathrm{ad}(H)|_{\mathfrak{n}_0}) \qquad (H \in \mathfrak{a}_0).$$

For a smooth admissible representation σ of M and $\nu \in \mathfrak{a}^*$, use the normalization

$$f(gman) = e^{-(\nu+\rho_P)(\log a)}\sigma(m)^{-1}f(g), \qquad (\pi_\nu(g_0)f)(g) = f(g_0^{-1}g). \tag{3.2}$$

The common compact realization of $I_P(\sigma, \nu) \otimes S$ consists of K-finite functions $F : K \to V_\sigma \otimes S$ satisfying

$$F(km) = (\sigma(m)^{-1} \otimes 1)F(k) \qquad (m \in K \cap M). \tag{3.3}$$

The spin factor is external to the induced representation: in (3.2), $\sigma(m)$ acts only on V_σ. If the inducing datum is given as a finite-length Harish-Chandra module, choose a smooth admissible globalization and take the K-finite vectors of the induced representation. Let $s_S(\widetilde{k}_0)$ denote the action on S of the Clifford component of $\widetilde{k}_0 \in \widetilde{K}$, lying over $k_0 \in K$. The diagonal action on the compact realization is

$$(\widetilde{k}_0 \cdot F)(k) = (1 \otimes s_S(\widetilde{k}_0))F(k_0^{-1}k).$$

Its finite vectors agree with the vectors finite under external left translation by K, because S is finite dimensional. Write D_ν for the Dirac operator in this realization.

Lemma 3.5 (Anticommutator identity)**.** *For $H \in \mathfrak{a}_0$, define*

$$(C_H F)(k) = c(\mathrm{Ad}(k)H)F(k).$$

This operator preserves the compact realization and its K-finite vectors. Moreover,

$$D_\nu C_H + C_H D_\nu = -2\nu(H)\,\mathrm{id}\,. \tag{3.4}$$

Proof. Since M centralizes $\mathfrak{a}$, multiplication by $c(\mathrm{Ad}(k)H)$ preserves (3.3). More explicitly, let $\widehat{F}$ be the extension of F to G determined by (3.2). For $g = kman$, the induced section corresponding to $C_H F$ is

$$\widehat{C_H F}(g) = c(\mathrm{Ad}(k)H)\widehat{F}(g). \tag{3.5}$$

Changing the compact factor k to km_0, with $m_0 \in K \cap M$, does not change $\mathrm{Ad}(k)H$. Clifford multiplication commutes with the inducing action on V_σ, so (3.5) is well defined and has the required right transformation law. Its scalar coefficient functions belong to a finite-dimensional space of matrix coefficients of the adjoint representation of K. Multiplication by these functions preserves K-finiteness.

Let κ be projection onto $\mathfrak{k}_0$ in the direct-sum decomposition

$$\mathfrak{g}_0 = \mathfrak{k}_0 \oplus (\mathfrak{m}_0 \cap \mathfrak{p}_0) \oplus \mathfrak{a}_0 \oplus \mathfrak{n}_0.$$

Fix $k \in K$, and choose a B-orthonormal basis $Z_1, \dots, Z_d$ of $\mathfrak{p}_0$. Then $\mathrm{Ad}(k)Z_1, \dots, \mathrm{Ad}(k)Z_d$ is also such a basis. Locally write

$$\exp(-tZ_j) = k_j(t)m_j(t)a_j(t)n_j(t),$$

choosing the tangent of $m_j(t)$ in $\mathfrak{m}_0 \cap \mathfrak{p}_0$. Then $k_j'(0) = \kappa(-Z_j)$ and

$$\exp(-t\,\mathrm{Ad}(k)Z_j)k = kk_j(t)m_j(t)a_j(t)n_j(t).$$

Differentiation of (3.2) and (3.5), followed by the product rule, yields

$$\begin{aligned}&\big(\pi_\nu(\mathrm{Ad}(k)Z_j)(C_H F)\big)(k)\\ &\quad = c\big(\mathrm{Ad}(k)[\kappa(-Z_j), H]\big)F(k) + c(\mathrm{Ad}(k)H)\big(\pi_\nu(\mathrm{Ad}(k)Z_j)F\big)(k).\end{aligned} \tag{3.6}$$

The inducing action of σ commutes with Clifford multiplication because the two actions are on distinct tensor factors.

Applying the Clifford relations to (3.6) gives

$$\begin{aligned}\big((D_\nu C_H + C_H D_\nu)F\big)(k) &= \sum_j c(\mathrm{Ad}(k)Z_j)c\big(\mathrm{Ad}(k)[\kappa(-Z_j), H]\big)F(k)\\ &\quad - 2\big(\pi_\nu(\mathrm{Ad}(k)H)F\big)(k).\end{aligned} \tag{3.7}$$

Since $\exp(-t\,\mathrm{Ad}(k)H)k = k\exp(-tH)$, the second term is

$$-2(\nu + \rho_P)(H)F(k). \tag{3.8}$$

The first term in (3.7) is independent of the chosen orthonormal basis: it is the image of the inverse bilinear-form tensor $\sum_j Z_j \otimes Z_j \in \mathfrak{p}_0 \otimes \mathfrak{p}_0$ under the bilinear map

$$U \otimes V \longmapsto c(\mathrm{Ad}(k)U)c\big(\mathrm{Ad}(k)[\kappa(-V), H]\big).$$

To evaluate this term, decompose $\mathfrak{n}_0$ into its nonzero $\mathfrak{a}_0$-weight spaces. For each weight α occurring in $\mathfrak{n}_0$, choose a B_θ-orthonormal basis $X_{\alpha,1}, \dots, X_{\alpha,m_\alpha}$ and set

$$Z_{\alpha,r} = \frac{X_{\alpha,r} - \theta X_{\alpha,r}}{\sqrt{2}}, \qquad T_{\alpha,r} = \frac{X_{\alpha,r} + \theta X_{\alpha,r}}{\sqrt{2}}.$$

The vectors $Z_{\alpha,r}$, together with orthonormal bases of $\mathfrak{m}_0 \cap \mathfrak{p}_0$ and $\mathfrak{a}_0$, form an orthonormal basis of $\mathfrak{p}_0$. Furthermore,

$$-Z_{\alpha,r} = T_{\alpha,r} - \sqrt{2}X_{\alpha,r}, \qquad \kappa(-Z_{\alpha,r}) = T_{\alpha,r}, \qquad [T_{\alpha,r}, H] = -\alpha(H)Z_{\alpha,r}.$$

It follows that

$$c(\mathrm{Ad}(k)Z_{\alpha,r})c\big(\mathrm{Ad}(k)[T_{\alpha,r},H]\big) = \alpha(H)\,\mathrm{id}\,.$$

For basis vectors in $(\mathfrak{m}_0 \cap \mathfrak{p}_0) \oplus \mathfrak{a}_0$, the compact projection is zero. Consequently the first term in (3.7) equals

$$\left(\sum_\alpha m_\alpha \alpha(H)\right) F(k) = 2\rho_P(H)F(k).$$

Combining this equality with (3.8) proves (3.4) at every $k \in K$. □

Corollary 3.6. *If $\nu \neq 0$, then*

$$H_D(I_P(\sigma,\nu)) = 0.$$

Proof of Theorem 1.1 and Corollary 3.6. Since $\mathfrak{a}_0$ spans $\mathfrak{a}$ over $\mathbb{C}$, there exists $H \in \mathfrak{a}_0$ with $\nu(H) \neq 0$. Apply Lemma 2.1 to the identity in Lemma 3.5, with $a = \nu(H)$. □

Remark 3.7 (Dependence on the inducing exponent). Let $\nu^\sharp \in \mathfrak{a}$ be defined by $B(\nu^\sharp, H) = \nu(H)$ for $H \in \mathfrak{a}$. Differentiating the inducing character gives

$$D_\nu = D_0 + C_{\nu^\sharp}.$$

Indeed, at $k \in K$, the difference $\pi_\nu(Z) - \pi_0(Z)$ is multiplication by $\nu(\mathrm{pr}_{\mathfrak{a}}(\mathrm{Ad}(k^{-1})Z))$. The projection onto $\mathfrak{a}$ restricts on $\mathfrak{p}$ to the B-orthogonal projection, so contraction with the Clifford factor is $c(\mathrm{Ad}(k)\nu^\sharp)$. The pointwise Clifford relations and Lemma 3.5 at $\nu = 0$ give

$$D_\nu^2 = D_0^2 - B(\nu^\sharp,\nu^\sharp)\,\mathrm{id}\,.$$

If σ is unitary, then $H_D(I_P(\sigma,0)) = \ker D_0$. For $\nu \neq 0$, Corollary 3.6 already gives vanishing, even when $B(\nu^\sharp,\nu^\sharp) = 0$.

Remark 3.8. For $\widetilde{k} \in \widetilde{K}$ lying over $k \in K$, put $\epsilon(k) = \det(\mathrm{Ad}(k)|_{\mathfrak{p}_0})$. The Pin convention gives

$$\widetilde{\pi}(\widetilde{k})D_\nu\widetilde{\pi}(\widetilde{k})^{-1} = \epsilon(k)D_\nu, \qquad \widetilde{\pi}(\widetilde{k})C_H\widetilde{\pi}(\widetilde{k})^{-1} = \epsilon(k)C_H.$$

The proof of Lemma 3.5 is a pointwise calculation and therefore applies also when K has orientation-reversing elements.

3.4. **The twisted index and Dirac cohomology of standard modules.** Let $Y = I_P(\sigma,\nu)$ be a standard $(\mathfrak{g},K)$-module as in Definition 3.3, with the notation introduced in Subsection 3.2. By Corollary 3.6 and the preceding subsection, it suffices to consider $\nu = 0$, in which case Y is tempered. We have the following:

Theorem 3.9 (Dong–Huang)**.** *Let Y be an irreducible tempered $(\mathfrak{g},K)$-module. Then $H_D(Y) \neq 0$ if and only if there exist a θ-stable Borel subalgebra $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{u}$, with fundamental Cartan subgroup $H = TA$, and a unitary character λ of H such that*

$$Y \simeq A_{\mathfrak{b}}(\lambda) = X_{\mathfrak{b}}(\lambda), \qquad \theta\lambda = \lambda,$$

and the following conditions hold:

(i) *The character λ is* weakly good*, meaning that*

$$\mathrm{Re}\langle \lambda + \rho(\mathfrak{u}), \alpha^\vee\rangle \geq 0 \qquad (\alpha \in \Delta(\mathfrak{u},\mathfrak{h})).$$

(ii) *Relative to the positive roots of $(\mathfrak{k},\mathfrak{t})$ determined by $\mathfrak{u} \cap \mathfrak{k}$,*

$$\lambda|_{\mathfrak{t}} + \rho(\mathfrak{u} \cap \mathfrak{p}) \quad \textit{is dominant for } \mathfrak{k}. \tag{3.9}$$

Proof of Theorem 1.2. Let $Y = I_P(\sigma,\nu)$ be a standard $(\mathfrak{g},K)$-module. Suppose that $H_D(Y) \neq 0$. By Corollary 3.6, $\nu = 0$, so Y is irreducible tempered. Theorem 3.9 gives $Y \simeq X_{\mathfrak{b}}(\lambda)$ with $\theta\mathfrak{b} = \mathfrak{b}$ and $\theta\lambda = \lambda$. Theorem 3.4 therefore implies $I_\theta(Y^\varepsilon) \neq 0$, so Y is twisted-elliptic. Conversely, a nonzero twisted Dirac index implies nonzero Dirac cohomology by definition. □

4. Application to graded affine Hecke algebras

We apply the results in Section 2 to parabolically induced module for graded affine Hecke algebras. The resulting vanishing for a nonzero inducing character is the counterpart of Theorem 1.1. The geometric application below recovers the non-tempered vanishing in Chan's theorem.

4.1. **Definitions and normalization.** Let $V_{\mathbb{R}}$ be a nonzero finite-dimensional real Euclidean space with inner product B, and let $R \subset V_{\mathbb{R}}$ be a reduced root system spanning $V_{\mathbb{R}}$. Let W be its Weyl group. Choose simple roots Π and positive roots R^+. Put

$$V = V_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}, \qquad \alpha^\vee = \frac{2\alpha}{B(\alpha,\alpha)}.$$

The form B is extended complex bilinearly. These spaces and this form are independent of the Lie-algebra notation in Section 3.

Fix a W-invariant function $k : R \to \mathbb{C}$. The graded affine Hecke algebra $\mathbb{H}$ is generated by $S(V)$ and $\mathbb{C}[W]$, with standard group elements t_w, subject to

$$t_{s_\alpha} v - s_\alpha(v) t_{s_\alpha} = k_\alpha B(v, \alpha^\vee) \qquad (\alpha \in \Pi,\ v \in V). \tag{4.1}$$

Its Poincaré–Birkhoff–Witt decomposition is $\mathbb{H} \simeq \mathbb{C}[W] \otimes S(V)$ as a vector space. For a nonreduced presentation, choose one reduced root system with the same reflection hyperplanes and use the coefficients determined by (4.1). More precisely, replacing a simple root α by $b_\alpha \alpha$, with $b_\alpha > 0$, replaces k_α by $b_\alpha k_\alpha$. The product $k_\alpha \alpha^\vee$ is unchanged. All sums below are over the positive roots of the chosen reduced system, with one term for each reflection.

For $v \in V$, define

$$\widetilde{v} = v - A(v), \qquad A(v) = \frac{1}{2} \sum_{\alpha \in R^+} k_\alpha B(v, \alpha^\vee) t_{s_\alpha}. \tag{4.2}$$

Let $C(V)$ be the Clifford algebra with relations

$$c(v)c(u) + c(u)c(v) = -2B(v,u)\,\mathrm{id} \qquad (v, u \in V),$$

and choose an irreducible $C(V)$-module $S_{\mathbb{H}}$. For a B-orthonormal basis $e_1, \dots, e_n$ of V, put

$$D_{\mathbb{H}} = \sum_{i=1}^{n} \widetilde{e}_i \otimes c(e_i). \tag{4.3}$$

This element is independent of the orthonormal basis. For an $\mathbb{H}$-module E, define

$$H_D(E) = \ker D_{\mathbb{H}} / (\ker D_{\mathbb{H}} \cap \operatorname{im} D_{\mathbb{H}}),$$

where $D_{\mathbb{H}}$ acts on $E \otimes S_{\mathbb{H}}$. The modified generators are those of [2, (2.13)], and (4.3) is the Dirac element of [2, Definition 3.1], with the root-space convention of [4, §2.3]. The polynomial space denoted by $V^\vee$ in [2] corresponds to V here after identifying the dual Euclidean spaces.

4.2. **Parabolic induction and the anticommutator.** For $J \subseteq \Pi$, let W_J be generated by $\{s_\alpha : \alpha \in J\}$, and let $\mathbb{H}_J$ be the subalgebra generated by all of $S(V)$ and $\mathbb{C}[W_J]$. Define

$$\mathfrak{a}_J = V^{W_J}.$$

Every $H \in \mathfrak{a}_J$ is central in $\mathbb{H}_J$: it commutes with $S(V)$, and (4.1) gives $t_{s_\alpha} H = H t_{s_\alpha}$ for $\alpha \in J$.

Lemma 4.1 ([2, Proposition 2.10(2)])**.** *For every $v \in V$ and $w \in W$,*

$$\widetilde{v}\, t_w = t_w \widetilde{w^{-1} v}.$$

Proof. For $\alpha \in \Pi$, the reflection s_α permutes $R^+ \setminus \{\alpha\}$. The definition of $A(v)$ gives

$$t_{s_\alpha} A(v) t_{s_\alpha} = A(s_\alpha v) + k_\alpha B(v, \alpha^\vee) t_{s_\alpha}.$$

Subtract this identity from

$$t_{s_\alpha} v t_{s_\alpha} = s_\alpha v + k_\alpha B(v, \alpha^\vee) t_{s_\alpha}$$

to obtain $t_{s_\alpha} \widetilde{v}\, t_{s_\alpha} = \widetilde{s_\alpha v}$. The assertion follows because the simple reflections generate W. □

Proposition 4.2 (Hecke algebra anticommutator)**.** *Let U be an $\mathbb{H}_J$-module and put $E = \mathbb{H}\otimes_{\mathbb{H}_J} U$. For $H \in \mathfrak{a}_J$, define*

$$T_H(h \otimes u \otimes s) = h \otimes Hu \otimes s \qquad (h \in \mathbb{H},\ u \in U,\ s \in S_{\mathbb{H}}).$$

This is a well-defined endomorphism of $E\otimes S_{\mathbb{H}}$ commuting with $D_{\mathbb{H}}$. In the vector-space realization

$$E \simeq \mathbb{C}[W] \otimes_{\mathbb{C}[W_J]} U,$$

the formula

$$C_H(t_w \otimes u \otimes s) = t_w \otimes u \otimes c(wH)s \tag{4.4}$$

defines an endomorphism of $E \otimes S_{\mathbb{H}}$ satisfying

$$D_{\mathbb{H}}C_H + C_H D_{\mathbb{H}} = -2T_H. \tag{4.5}$$

If H acts invertibly on U, then $H_D(E) = 0$. In particular, if $H|_U = a\,\mathrm{id}_U$ for $a \neq 0$, then $D_{\mathbb{H}}C_H + C_H D_{\mathbb{H}} = -2a\,\mathrm{id}$ and $H_D(E) = 0$.

Proof. Since H is central in $\mathbb{H}_J$, the endomorphism $u \mapsto Hu$ is $\mathbb{H}_J$-linear. It therefore induces an $\mathbb{H}$-linear endomorphism of E. Acting trivially on the spin factor, this endomorphism is T_H, which consequently commutes with $D_{\mathbb{H}}$. The stated realization follows from the Poincaré–Birkhoff–Witt decomposition. If $j \in W_J$, then

$$t_{wj} \otimes u = t_w \otimes t_j u, \qquad wjH = wH.$$

Consequently (4.4) respects the relations in the balanced tensor product and is well defined.

Lemma 4.1 gives, for every $w \in W$,

$$D_{\mathbb{H}}(t_w \otimes u \otimes s) = \sum_i t_w \otimes e_i u \otimes c(we_i)s - \frac{1}{2}\sum_{\alpha\in R^+} k_\alpha t_{ws_\alpha} \otimes u \otimes c(w\alpha^\vee)s.$$

Indeed, use the orthonormal basis (we_i) in (4.3), move t_w past each modified generator, and expand (4.2).

The first sum in (4.6) contributes to the anticommutator the expression

$$\begin{aligned}\sum_i t_w \otimes e_i u \otimes \big(c(we_i)c(wH) + c(wH)c(we_i)\big)s &= -2\sum_i B(e_i, H)t_w \otimes e_i u \otimes s \\ &= -2t_w \otimes Hu \otimes s = -2T_H(t_w \otimes u \otimes s).\end{aligned}$$

For each $\alpha \in R^+$, its reflection term contributes a scalar multiple of

$$c(w\alpha^\vee)c(wH) + c(ws_\alpha H)c(w\alpha^\vee). \tag{4.6}$$

This expression is zero. To verify the assertion, first observe that

$$s_\alpha H = H - B(H, \alpha^\vee)\alpha, \qquad c(\alpha)c(\alpha^\vee) = -2\,\mathrm{id}\,.$$

It follows that

$$c(\alpha^\vee)c(H) + c(s_\alpha H)c(\alpha^\vee) = -2B(\alpha^\vee, H)\,\mathrm{id} - B(H, \alpha^\vee)c(\alpha)c(\alpha^\vee) = 0.$$

Since w preserves B, the same equality after applying w proves the vanishing of (4.6). Thus every reflection term cancels separately, proving (4.5). If $H|_U$ is invertible, its inverse is $\mathbb{H}_J$-linear, and

$$T_H^{-1} = 1 \otimes (H|_U)^{-1} \otimes 1.$$

Lemma 2.1 applies. Explicitly,

$$x = D_{\mathbb{H}}\left(-\tfrac{1}{2}C_H T_H^{-1} x\right) \qquad (x \in \ker D_{\mathbb{H}}).$$

The scalar case follows by taking $T_H = a\,\mathrm{id}$. □

4.3. **Vanishing for a nonzero inducing character.** For $J \subseteq \Pi$, put $V_J = \operatorname{span}_{\mathbb{C}} J$. Then $V = V_J \oplus \mathfrak{a}_J$ orthogonally. Let $\mathbb{H}_J^{\mathrm{ss}}$ be generated by $S(V_J)$ and $\mathbb{C}[W_J]$. The cross relations and the Poincaré–Birkhoff–Witt decomposition give

$$\mathbb{H}_J \simeq \mathbb{H}_J^{\mathrm{ss}} \otimes S(\mathfrak{a}_J).$$

For $\nu \in \mathfrak{a}_J^*$, let $\mathbb{C}_\nu$ be the one-dimensional $S(\mathfrak{a}_J)$-module on which H acts by $\nu(H)$. For an $\mathbb{H}_J^{\mathrm{ss}}$-module σ, put

$$E(J, \sigma, \nu) = \mathbb{H} \otimes_{\mathbb{H}_J} (\sigma \otimes \mathbb{C}_\nu).$$

Corollary 4.3 (Nonzero inducing character)**.** *For every $J \subseteq \Pi$, every $\mathbb{H}_J^{\mathrm{ss}}$-module σ, and every $\nu \in \mathfrak{a}_J^* \setminus \{0\}$,*

$$H_D(E(J, \sigma, \nu)) = 0.$$

The parameter function k may be complex-valued.

Proof. Choose $H \in \mathfrak{a}_J$ with $\nu(H) \neq 0$. On $\sigma \otimes \mathbb{C}_\nu$, the element H acts by the scalar $\nu(H)$. Proposition 4.2 gives

$$D_{\mathbb{H}} C_H + C_H D_{\mathbb{H}} = -2\nu(H)\,\mathrm{id}\,.$$

Lemma 2.1 proves the assertion. □

Definition 4.4. Let $k \in \mathbb{R}$, and let $\omega_\alpha^J \in V_J$ be characterized by $B(\omega_\alpha^J, \beta^\vee) = \delta_{\alpha\beta}$ for $\alpha, \beta \in J$. A finite-dimensional $\mathbb{H}_J^{\mathrm{ss}}$-module σ is *tempered* if every generalized $S(V_J)$-weight $\eta \in V_J^*$ of σ satisfies

$$\operatorname{Re}\eta(\omega_\alpha^J) \leq 0 \qquad (\alpha \in J).$$

A *standard module* (in the sense of [10, §3, Theorem 3.3] for instance) is of the form $E(J, \sigma, \nu)$, where σ is irreducible tempered and $\nu \in \mathfrak{a}_J^*$, extended by zero on V_J, satisfies

$$\operatorname{Re}\nu(\alpha) > 0 \qquad (\alpha \in \Pi \setminus J).$$

Corollary 4.5. *Retain the setting in the above definition, and let $E = E(J, \sigma, \nu)$ be a Langlands standard module. If $J \neq \Pi$, then $H_D(E) = 0$. Consequently every non-tempered Langlands standard module has zero Dirac cohomology.*

Proof. If $J \neq \Pi$, the strict chamber condition implies $\nu \neq 0$, so Corollary 4.3 applies. If $J = \Pi$, then $\mathfrak{a}_J = V^W = 0$, and $E = \sigma$ is tempered by definition. □

Remark 4.6. For crystallographic R, the ordinary Dirac index defined in [5, §2.5] vanishes for every finite-dimensional module induced from a proper parabolic subalgebra; see [5, §2.10 and Remark 3.1]. This is an equality of virtual representations and does not by itself imply vanishing of Dirac cohomology. Corollary 4.3 proves the latter vanishing when $\nu \neq 0$.

4.4. **Geometric realizations and parabolic induction.** We now apply Proposition 4.2 to the *geometric standard module* $E_{s,e,r,\zeta}$ constructed in [4, Definition 4.6] (up to a positive scaling). Let $\mathcal{G}$ denote the connected complex reductive group defining the geometric parameters, and let $\mathfrak{h}_{\mathrm{geom}}$ be the Lie algebra of the connected centre of the cuspidal Levi subgroup. Thus $V = \mathfrak{h}_{\mathrm{geom}}^*$. Here e is nilpotent and s is semisimple in the Lie algebra of $\mathcal{G}$, $[s, e] = 2re$, and ζ is an irreducible representation of the component group specified there.

Explicitly, put

$$\mathcal{Z}_{\mathcal{G}}(e, s) = \{(g, a) \in \mathcal{G} \times \mathbb{C}^\times : \mathrm{Ad}(g)e = a^2 e,\ \mathrm{Ad}(g)s = s\},$$
$$A_{\mathcal{G}}(e, s) = \mathcal{Z}_{\mathcal{G}}(e, s)/\mathcal{Z}_{\mathcal{G}}(e, s)^\circ.$$

Here the superscript $\circ$ denotes the identity component, so $A_{\mathcal{G}}(e, s)$ is a finite group. Its action on the geometric module $E_{s,e,r}$ commutes with the $\mathbb{H}$-action, and ζ is an irreducible complex representation of this group. The module with parameter ζ is the multiplicity space

$$E_{s,e,r,\zeta} = \mathrm{Hom}_{A_{\mathcal{G}}(e,s)}(\zeta, E_{s,e,r}),$$

as in [4, Definition 4.6]. We write $E_{s,e,r,\zeta}^{\mathcal{G}}$ when it is necessary to specify the ambient group.

Throughout this subsection, assume $r \in \mathbb{R} \setminus \{0\}$. Let $\omega_\alpha \in V$ be the fundamental weights, characterized by $B(\omega_\alpha, \beta^\vee) = \delta_{\alpha\beta}$. Following [4, Definition 4.5], $E_{s,e,r,\zeta}$ is *tempered* if every generalized $S(V)$-weight $\lambda \in V^*$ of $E_{s,e,r,\zeta}$ satisfies

$$\frac{\operatorname{Re}\lambda(\omega_\alpha)}{r} \geq 0 \qquad (\alpha \in \Pi).$$

Proposition 4.7. *Let $E = E^{\mathcal{G}}_{s,e,r,\zeta}$ be a nonzero geometric standard module which is not tempered. Choose an $\mathfrak{sl}_2$-triple (e, h_e, f_e) satisfying $[s, h_e] = 0$ and $[s, f_e] = -2rf_e$, and put $z = s - rh_e$. For $c \in \mathbb{C}$, set*

$$\mathfrak{g}_{\mathrm{geom}}(c) = \{x \in \mathrm{Lie}(\mathcal{G}) : [z, x] = cx\}.$$

Let Q be the parabolic subgroup with Levi subgroup $\mathcal{M}$ defined by

$$\mathrm{Lie}(Q) = \bigoplus_{\mathrm{Re}(c)/r \leq 0} \mathfrak{g}_{\mathrm{geom}}(c), \qquad \mathrm{Lie}(\mathcal{M}) = \bigoplus_{\mathrm{Re}(c)/r = 0} \mathfrak{g}_{\mathrm{geom}}(c). \tag{4.7}$$

The inherited cuspidal datum identifies the Hecke algebra of $\mathcal{M}$ with $\mathbb{H}_J$ for some $J \subsetneq \Pi$. Let $\zeta_{\mathcal{M}}$ correspond to ζ under the natural isomorphism of the component groups for (e, s) in $\mathcal{M}$ and $\mathcal{G}$. Then the geometric standard module for $\mathcal{M}$ with the same s, e, r satisfies

$$U = E^{\mathcal{M}}_{s,e,r,\zeta_{\mathcal{M}}}, \qquad E^{\mathcal{G}}_{s,e,r,\zeta} \simeq \mathbb{H} \otimes_{\mathbb{H}_J} E^{\mathcal{M}}_{s,e,r,\zeta_{\mathcal{M}}}.$$

Moreover, U is irreducible and has the form

$$U \simeq \sigma \otimes \mathbb{C}_\nu, \qquad \nu \in \mathfrak{a}_J^* \setminus \{0\},$$

where σ is an irreducible $\mathbb{H}_J^{\mathrm{ss}}$-module. In particular, some $H_0 \in \mathfrak{a}_J$ acts invertibly on U.

Proof. The compatible triple and the parabolic subgroup in (4.7) are constructed in [9, §§3.11–3.12]. For this Levi subgroup, Lusztig's induction theorem gives induction from the nonzero irreducible module U, as recalled in [4, §5.1]. The identification of the component groups and the formula $U = E^{\mathcal{M}}_{s,e,r,\zeta_{\mathcal{M}}}$ are supplied by [9, §3.38(c),(e) and §3.39]; the inherited algebra is the full parabolic algebra $\mathbb{H}_J$ by [9, §1.16]. Since $S(\mathfrak{a}_J)$ is central in $\mathbb{H}_J$, Schur's lemma gives $U \simeq \sigma \otimes \mathbb{C}_\nu$ with σ irreducible. It remains to prove that $J \neq \Pi$ and $\nu \neq 0$.

Choose a parabolic subgroup P_0 in the fixed cuspidal conjugacy class such that $P_0 \subseteq Q$ and $z \in \mathfrak{p}_0 = \mathrm{Lie}(P_0)$, and put

$$t = z \bmod [\mathfrak{p}_0, \mathfrak{p}_0] \in \mathfrak{p}_0/[\mathfrak{p}_0, \mathfrak{p}_0] \simeq \mathfrak{h}_{\mathrm{geom}}.$$

Such P_0 exists, and t is independent of its choice, by [9, Lemma 3.25]. Lemma 3.26(a),(b) gives

$$\frac{\operatorname{Re}\alpha(t)}{r} \begin{cases} = 0, & \alpha \in J, \\ < 0, & \alpha \in \Pi \setminus J. \end{cases} \tag{4.8}$$

Moreover, Lemma 3.26(c),(d), together with the weight identifications in [9, §3.36(c),(d) and §3.38(d)], gives, for every generalized $S(V)$-weight η of U,

$$\frac{\operatorname{Re}\eta(\omega_\alpha)}{r} \geq \frac{\operatorname{Re}\omega_\alpha(t)}{r} \qquad (\alpha \in \Pi), \tag{4.9}$$

with equality for $\alpha \notin J$. These statements use our fundamental weights, which are positive multiples of Lusztig's dominant generators [9, §§3.4–3.6].

If $J = \Pi$, then $Q = \mathcal{G}$ and $U = E$. Since the roots span V, (4.8) implies $\operatorname{Re}\omega_\alpha(t)/r = 0$ for every α. Equation (4.9) would then make E tempered. Thus $J \neq \Pi$.

Choose $\beta \in \Pi \setminus J$. Then $H_0 = \omega_\beta \in \mathfrak{a}_J$, and every weight of U restricts to ν on $\mathfrak{a}_J$. Write

$$\omega_\beta = \sum_{\alpha \in \Pi} b_\alpha \alpha, \qquad b_\alpha \geq 0, \qquad b_\beta > 0,$$

using positivity of the inverse Cartan matrix on each irreducible component. The equality case of (4.9) and (4.8) yield

$$\frac{\operatorname{Re}\nu(H_0)}{r} = \frac{\operatorname{Re}\omega_\beta(t)}{r} = \sum_{\alpha \in \Pi} b_\alpha \frac{\operatorname{Re}\alpha(t)}{r} < 0.$$

The β-summand is strictly negative and all other summands are nonpositive. Hence $\nu \neq 0$ and $H_0|_U = \nu(H_0)\,\mathrm{id}_U$ is invertible. □

Corollary 4.8 (Non-tempered geometric standard modules)**.** *For every non-tempered geometric standard module E,*

$$H_D(E) = 0.$$

Proof. Apply Proposition 4.2 to the induction description in Proposition 4.7. □

Remark 4.9. Corollary 4.8 gives an anticommutator proof of the non-tempered vanishing in [4, Theorem 6.4]. The tempered results and the twisted-index criterion are [4, Theorems 6.2–6.3 and Corollary 6.6].

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